

Assignment 2: Data and methodology

Thomas Hoang / Student ID: A1984077

Abstract

Insert abstract text here. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

1. Introduction & Related Work

This demo file is intended to serve as a “starter file”. It is for preparing manuscript submission only, not for preparing camera-ready versions of manuscripts. Manuscripts will be typeset for publication by the journal, after they have been accepted.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

2. Data Preparation & Pipeline Design

The dataset is taken from the internal Staging database at my organization (our internal sales calls using this system). I selected rows with SQL and exported the result to CSV.

```
SELECT *
FROM call_call
WHERE type='MANUAL'
      AND record != ''
      AND duration > 5
ORDER BY created_at DESC;
```

duration > 5 excludes empty or very short calls. The export contains 13,232 records.

shape: (13,232, 10)

call_id	phone	carrier	recording_url	duration_sec	start_at_raw	end_at_raw	call_status	hangup_cause	direction
i64	str	str	str	i64	str	str	str	str	str
770590	"*84876818134"	"OTHER"	"https://storage004.uccall.vn/..."	142	"2026-04-06 02:20:09.000000 +00..."	"2026-04-06 02:23:07.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770585	"*84943118148"	"VINAPHONE"	"https://storage004.uccall.vn/..."	13	"2026-04-04 10:13:10.000000 +00..."	"2026-04-04 10:13:45.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770584	"*84943118148"	"VINAPHONE"	"https://storage004.uccall.vn/..."	13	"2026-04-04 10:13:10.000000 +00..."	"2026-04-04 10:13:45.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770583	"*84343438000"	"VIETTEL"	"https://storage004.uccall.vn/..."	123	"2026-04-04 09:53:08.000000 +00..."	"2026-04-04 09:55:21.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
770582	"*84989524490"	"VIETTEL"	"https://storage004.uccall.vn/..."	249	"2026-04-04 09:28:51.000000 +00..."	"2026-04-04 09:33:13.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
...	...	...	...	...	...	...	...	...	...
7	"*84827903789"	"VINAPHONE"	"https://storage004.uccall.vn/..."	178	"2025-06-02 08:17:08.000000 +00..."	"2025-06-02 08:20:10.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
6	"*84975583171"	"VIETTEL"	"https://storage004.uccall.vn/..."	677	"2025-06-02 08:07:22.000000 +00..."	"2025-06-02 08:18:35.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
5	"*84978954696"	"VIETTEL"	"https://storage004.uccall.vn/..."	304	"2025-06-02 08:11:06.000000 +00..."	"2025-06-02 08:16:31.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"
4	"*84968982335"	"VIETTEL"	"https://storage004.uccall.vn/..."	24	"2025-06-02 08:12:46.000000 +00..."	"2025-06-02 08:13:31.000000 +00..."	"ANSWERED"	"NORMAL_CLEARING"	"OUTBOUND"

Mode: Command | Ln 1, Col 1 | 01_PrepDataCalls.ipynb

Figure 1. Sample rows from the exported call dataframe.

2.1 Data Visualization

Figure 2 plots `duration_sec` for all 13,232 calls with valid duration. The distribution is right-skewed: **most calls are short**. Durations above the 99th percentile (about 775 s) are clipped and grouped in the final bin so the main mass of the histogram remains visible.

In a call, the silence is often much longer than the speech. This **motivates segmenting audio** before speech-to-text instead of processing entire files end-to-end.

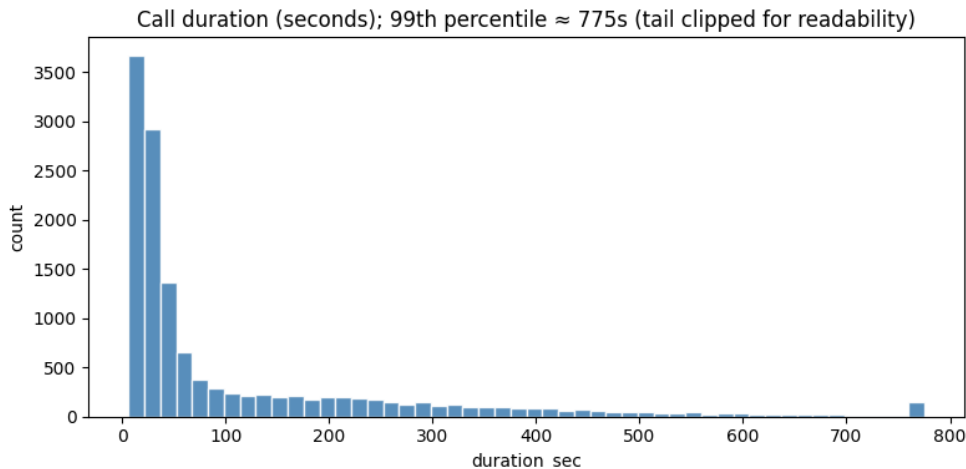


Figure 2. Distribution of call duration (seconds). Values above the 99th percentile are merged into the last bin.

2.2 Why do we need to preprocess the audio?

The goal is to feed a speech-to-text model **audio that is mostly speech**. Raw files add silence, hold tones, background noise, and non-speech events. That increases recognition errors. Preprocessing limits those segments before ASR.

Recordings are stereo: agent on the left channel and customer on the right in my setup. Downstream analysis targets the customer, so I take the right channel before VAD.

Voice activity detection (VAD) outputs time intervals with speech. I retain those intervals and drop the rest. That removes long silent regions and much background sound prior to ASR. It is an imperfect but standard front end.

2.3 Pre-processing Step 1: Voice Activity Detection (VAD)

I use **Silero VAD** (Silero Team 2024): low latency on CPU/GPU, suitable for batch jobs, and fast enough for real-time use.

Figure 3 is one example from my Silero VAD test. Left: both channels overlaid (labels as in my notebook). Right: customer channel only, with green spans for VAD-detected speech. Audio is resampled to **16 kHz** for the model.

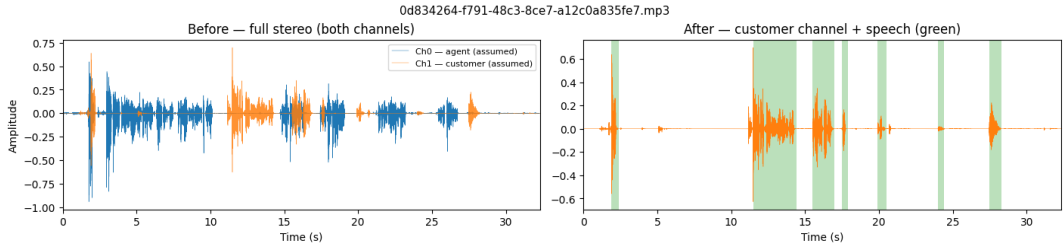


Figure 3. One call: stereo mix (left) and customer channel with VAD speech regions (right).

2.4 Pre-processing Step 2: Automatic Speech Recognition (ASR)

After VAD segmentation, each speech-only chunk is forwarded to an Automatic Speech Recognition (ASR) model. In this project, I use the Vietnamese wav2vec2 checkpoint `nguyenvulebinh/wav2vec2-base-vi Binh 2022`. It follows the wav2vec2 architecture and is self-supervised pre-trained on Vietnamese speech. The ASR model converts audio segments into unstructured text transcripts. Segment-level transcripts are then merged by timestamp to reconstruct each full-call transcript.

2.5 Step 3: LLM Fine-Tuning

In the final stage, reconstructed transcripts are turned into a supervised fine-tuning dataset for the downstream LLM. What the model must predict is not chosen ad hoc: it is fixed by a **system-level field configuration** (column names, types, allowed label sets, and output formats). That configuration is the single source of truth for acceptable response values at training and deployment time.

System field configuration.

```
column: customer_need_category
type: classification
groups:
  interested | not_interested
  | no_information | needs_more_information
  | price_or_budget_concern | timing_defer
  | comparing_options | already_satisfied
  | undecided
response: Enum[str]

column: customer_callback_time
type: optional_date
response: date | null
format: YYYY-MM-DD
notes:
  bare JSON null if unknown
```

For supervised fine-tuning, each transcript yields **one training row per configured field**. Each row is a JSON object with **instruction**, **input**, and **response**. The instruction is a natural-language prompt *instantiated from* that configuration: it states the role, the target field, and rules that enforce the configured types (for example, closed labels or date/null). The input is the transcript; the response is the supervised target and must be **consistent with the field definition above** (same enums, formats, and null policy). The following block shows two example rows (same transcript, two configured fields).

```
{
  "instruction": "You are a structured-field extractor for telesales QA. The user message
  ↳ contains one full call transcript (agent and customer).\n\nTask: assign
  ↳ customer_need_category.\n\nRules:\n- Return exactly one snake_case label from the closed
  ↳ set in customer_need_category.groups in the system configuration above.\n- Rely only on
  ↳ what the customer says or clearly implies; do not invent facts.\n- If the transcript
  ↳ does not support any label, return no_information.\n- Output the label token only: no
  ↳ quotes, commas, JSON, or explanation.",
  "input": "<transcript>",
  "response": "needs_more_information"
}
```

```
"instruction": "You are a structured-field extractor for telesales QA. The user message
↪ contains one full call transcript (agent and customer).\n\nTask: extract
↪ customer_callback_time.\n\nRules:\n- If the customer agrees to a callback and a calendar
↪ date is stated or unambiguously implied, return it as YYYY-MM-DD in Gregorian
↪ calendar.\n- If no callback is agreed or the date cannot be resolved, return JSON null
↪ (bare null, not a string).\n- Output only the date string or null, with no surrounding
↪ text.",
"input": "<transcript>",
"response": "2026-04-06"
}
```

3. Evaluation Metrics

The conclusion text goes here.

Notes

References

- Binh, Nguyen Vu Le. 2022. *Wav2vec2-base-vi*. <https://huggingface.co/nguyenvulebinh/wav2vec2-base-vi>. Vietnamese self-supervised pre-trained wav2vec2.
- Silero Team. 2024. *Silero vad: pre-trained enterprise-grade voice activity detector (vad), number detector and language classifier*. <https://github.com/snakers4/silero-vad>. GitHub repository.